SAVINDI WIJENAYAKA, Ph.D.

Senior Machine Learning & AI Engineer

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Summary

Senior Machine Learning & AI Engineer with a PhD and 6+ years of applied experience across traditional ML, Deep Learning, and Generative AI, developing and deploying systems in cloud-native environments. Brings technical depth across the ML lifecycle, from problem framing and research to model development, system design, and deployment, owning each phase with a focus on business impact, engineering rigour, and data governance.

Technical Skills

- **AI & ML:** LLM/VLM, RAG pipelines, Computer Vision, NLP, Signal Processing
- **Languages & Frameworks:** Python, FastAPI, PyTorch, Keras, Flask, Django, Java, Spring Boot
- **Data types:** Unstructured: Text, Image, Video, Audio, 3D, Point Cloud, Mesh; Structured: Time Series, Tabular
- **Cloud & Infrastructure:** Azure, GCP, AWS, Docker, Kubernetes, Data warehouse (Snowflake), Relational (MSSQL, Postgres), Vector & Embeddings (pgvector), Stream data (Azure Stream Analytics)
- **MLOps & Engineering:** CI/CD (GitHub Actions, Azure DevOps), Responsible AI & Governance, Git, REST, gRPC, Agile, Claude, Cursor

Experience

Machine Learning Engineer

Auror · Full-time | Sept 2025 - Present | Auckland, New Zealand

Auror is a retail crime intelligence platform that provides AI-enhanced capabilities to prevent crime.

- Spearheaded scoping, planning, and architecture of a RAG-based compliance questionnaire system on GCP, using a reranker, embeddings, Postgres with pgvector, and BAML for structured LLM outputs, driving cross-functional coordination across teams and directing two engineers through full-stack implementation.
- Designed a multi-gate evaluation framework for Auror's flagship entity resolution product, resolving class imbalance and circular bias; shipped a new model to production via a region-separated CI/CD pipeline, contributing to a 15-23pp increase in acceptance rates across all 4 regions.
- Architected a modular ML evaluation and fairness reporting framework for prospect trials, reducing report turnaround by 80% and accelerating the path to technical win.
- Established team-wide ML engineering standards via agentic coding conventions and a data licensing review, forming the foundation for a broader responsible AI and data governance framework.
- Prototyped an LPR camera health monitoring system by mapping vehicle detections to spatial heatmaps and flagging deviations from optimal detection zones, enabling automated identification of camera obstructions.

Machine Learning Engineer

WSO2 · Full-time | Sept 2020 - Nov 2021 | Colombo, Sri Lanka

WSO2 is a leading open-source integration vendor; Choreo is its AI-enhanced platform-as-a-service product.

- Designed and deployed an LLM-based generative AI system leveraging few-shot learning for automated API test case generation on Azure Kubernetes Service, reducing manual testing effort for customers and enabling an AI-assisted developer workflow.
- Co-architected a time series-based AI Anomaly Detection system using an event-driven microservices architecture across Azure Stream Analytics, Event Hubs, and Function Apps, enabling customers to detect abnormal application behaviours in real time.
- Implemented a regression testing pipeline for the Ballerina Language Server, systematically surfacing performance degradations and resource utilisation issues to ensure production reliability for users.

Software Research Engineer

Pearson · Internship Sept 2018 - Sept 2019 Colombo, Sri Lanka

Pearson is a leading global education provider offering curriculum materials, multimedia learning tools, and testing programs.

- Researched and built a Q&A chatbot for Pearson's educational content, iterating from BiDAF to fine-tuned BERT and GPT-2 models to improve answer quality and enable conversational querying of learning material.
- Researched and implemented deep learning modules for emotion detection and speech analysis, integrating models into scalable RESTful microservices for an AI Public Speaking Evaluator.
- Investigated and developed a flashcard classification service using ULMFiT, LSTM, and GRU to automatically categorise flashcards, reducing manual tagging effort and improving content discoverability for students.

Education

Ph.D. in Bioengineering

University of Auckland Dec 2021 - May 2025 Auckland, New Zealand

Pursued a Ph.D. specialising in Computer Vision-based deep learning for medical imaging and computational quantification, owning every phase from problem framing and dataset construction through model architecture, systematic experimentation, and pipeline delivery across 20+ distinct samples.

- Engineered a semantic segmentation model with multiscale attention mechanisms, conducting extensive ablation studies to optimise architecture and delivering an automated analysis pipeline that reduced manual processing time by 40+ hours per dataset.
- Developed a Python-based 3D quantification framework grounded in advanced mathematical methods, transforming raw micro-CT data into structured geometric benchmarks and standardising the full workflow from ground-truth labelling to downstream modelling and simulation.

B.Sc. (Hons.) in Software Engineering

University of Kelaniya Feb 2016 - Mar 2020 Kelaniya, Sri Lanka

- Specialised in Data Science and Net-centric application development.
- Attained a GPA of 3.96 out of 4.00, obtaining a First Class.
- Dean's List Honouree, recognised in all four academic years of the B.Sc. programme.

Knowledge Sharing & Technical Outreach

- Member of the teaching team for Code In Place 2021, an online Python course offered by Stanford University, contributing to global tech education initiatives.
- Served as a guest speaker for multiple technical webinars (organised by IEEE and DeepLearning.AI), effectively communicating complex deep learning topics to broader audiences.
- Authored technical articles on Medium covering conceptual topics (CNNs, Kubernetes internals, JVM), practical applications of cloud-native microservices (Kubernetes, Docker, Azure), and automated CI/CD pipelines (GitHub Actions, Azure ARM templates).

Achievements

- **1st Place (2022)** and **2nd Place (2024)** in the international SPARC FAIR Codeathon, representing the University of Auckland, organised by the SPARC Data and Resource Centre and the NIH.
- **4th Place** in DataStorm 2020 Datathon, organised by Octave (JKH) and University of Moratuwa.
- **1st Runner-Up** in National Youth Software Competition 2017, organised by UNDP Sri Lanka.
- **Vice President** of Marketing & Communications at AIESEC in the University of Kelaniya in 2018, contributing to local chapter growth.
- **Women's Karate Captain** at the University of Kelaniya, leading the team to national 3rd place while earning National and University Colours alongside multiple national and inter-university medals (2016–2018).